EM-S370(Z) / EM-S37B(Z)

High Reliability / Halogen Free

- Applications include: W/B, CSP, BGA, FC PKG, SiP, module and coreless.
- High heat resistance and excellent CAF resistance.
- Halogen free for IC packaging.
- RoHS Compliant
- UL File: E150504
- Applicable IPC Slash Sheets: IPC-4101 /127, /128, /130; IPC-4103 /250, /550

Basic Laminate Property

Property	Item		Typical Value	Unit	Test Condition	IPC-TM-650
Thermal	Tg		N/A	°C	DSC	2.4.25
			180	°C	TMA	2.4.24
			210	°C	DMA	2.4.24.4
	CTE, X/Y-axis		11/13	ppm/°C	< Tg, TMA	2.4.24.5
	CTE, Z-axis		35~40	ppm/°C	< Tg, TMA	2.4.24
			140~160	ppm/°C	> Tg, TMA	
	Td		390	°C	TGA (5% W.L)	2.4.24.6
	T288		>60	min.	Clad	2.4.24.1
			>60	min.	Etched	
	Thermal Conductivity		0.58	W/m.K	-	ASTM D5470
Electrical	Dk (RC: 50/70%)	1 GHz	4.4/4.0	-	C-24/23/50	2.5.5.9
		10 GHz	4.2/3.8	-		Cavity Resonator
	Df (RC: 50/70%)	1 GHz	0.012/0.014	-	C-24/23/50	2.5.5.9
		10 GHz	0.015/0.019	-		Cavity Resonator
	Volume Resistivity		$>10^{10}$	MΩ-cm	C-96/35/90	2.5.17.1
	Surface Resistivity		$>10^{9}$	MΩ	C-96/35/90	2.5.17.1
Physical	Water Absorption		0.14	%	E-1/105+D-24/23	2.6.2.1
	Peel Strength	VLP, 12 um	0.9	kg/cm	As Received	2.4.8
	Flexural Modulus	Warp	22~24	GPa	As Received	2.4.4
		Fill	20~23	GPa	As Received	
	Flame Resistance		V-0	-	C-48/23/50	UL-94

Above typical values are tested under specified constructions and not intended for specification.